

Identification and Control of a Crane System

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Abstract—This paper presents the experimental identification and control of a crane system using Integral State Feedback (ISF) techniques. Experimental data from the physical plant were processed using MATLAB System Identification Toolbox in order to derive empirical transfer functions for the cart position and pendulum angle dynamics.

The identified models were combined into a complete SIMO state-space representation suitable for modern control design. Two controllers were considered for the project: Integral State Feedback via pole placement and Integral State Feedback using the Linear Quadratic Regulator (LQR) criterion. Those controllers were evaluated according to settling time, overshoot, steady-state error, and pendulum oscillation attenuation.

I. INTRODUCTION

II. SYSTEM IDENTIFICATION

A. Experimental Setup

The experimental platform consists of a cart driven by a DC motor through a pulley-belt transmission system. A pendulum is attached to the cart and both cart position and pendulum angle are measured using incremental optical encoders.

The plant was excited using a trapezoidal voltage input of approximately 7.5 V. Experimental responses were sampled every 20 ms.

B. Position Transfer Functions

The identified cart-position transfer functions obtained from the selected experiments were:

$$G_{p1}(s) = \frac{0.2493}{s^2 + 10.41s + 7.838 \times 10^{-7}} \quad (1)$$

$$G_{p2}(s) = \frac{0.2542}{s^2 + 10.5s + 1.146 \times 10^{-6}} \quad (2)$$

$$G_{p5}(s) = \frac{0.2546}{s^2 + 10.44s + 8.605 \times 10^{-7}} \quad (3)$$

$$G_{p6}(s) = \frac{0.2611}{s^2 + 10.69s + 6.719 \times 10^{-7}} \quad (4)$$

Table I summarizes the extracted parameters.

TABLE I
POSITION MODEL PARAMETERS

Model	K	a_x
grua1	0.2493	10.41
grua2	0.2542	10.50
grua5	0.2546	10.44
grua6	0.2611	10.69

C. Angular Transfer Functions

The identified pendulum-angle transfer functions were:

$$G_{\theta 1}(s) = \frac{-0.3494s - 0.4978}{s^2 + 0.01909s + 35.34} \quad (5)$$

$$G_{\theta 2}(s) = \frac{-0.3951s - 0.3945}{s^2 + 0.01639s + 35.33} \quad (6)$$

$$G_{\theta 5}(s) = \frac{-0.4017s - 0.4771}{s^2 + 0.01663s + 35.36} \quad (7)$$

$$G_{\theta 6}(s) = \frac{-0.3969s - 0.5118}{s^2 + 0.01526s + 35.36} \quad (8)$$

The corresponding parameters are summarized in Table II.

TABLE II
ANGULAR MODEL PARAMETERS

Model	ω_n	ζ	K
grua1	5.9447	0.001606	-0.0141
grua2	5.9439	0.001379	-0.0112
grua5	5.9464	0.001398	-0.0135
grua6	5.9464	0.001283	-0.0145

D. Average SIMO Model

Average transfer functions were obtained from the identified models:

$$G_p(s) = \frac{0.2548}{s^2 + 10.51s + 8.66 \times 10^{-7}} \quad (9)$$

$$G_{\theta}(s) = \frac{-0.3858s - 0.4703}{s^2 + 0.01685s + 35.35} \quad (10)$$

Using the state vector:

$$x^T = [p \quad \theta \quad \dot{p} \quad \dot{\theta}] \quad (11)$$

the resulting SIMO state-space model becomes:

$$\dot{x} = Ax + Bu \quad (12)$$

with:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -10.51 & 0 \\ 0 & -35.35 & 0 & -0.0169 \end{bmatrix} \quad (13)$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0.2548 \\ -0.3858 \end{bmatrix} \quad (14)$$

$$C = [1 \ 0 \ 0 \ 0] \quad (15)$$

III. CONTROLLER DESIGN

An Integral State Feedback strategy was implemented to eliminate steady-state error while regulating the cart position.

The augmented system was constructed by adding an integral state associated with the position tracking error.

A. Pole Placement Design

B. LQR Design

For the LQR controller, the system was augmented with an integral state:

$$\dot{x}_I = r - C_r x \quad (16)$$

where:

$$C_r = [1 \ 0 \ 0 \ 0] \quad (17)$$

The augmented system matrices become:

$$A_{aug} = \begin{bmatrix} A & 0 \\ -C_r & 0 \end{bmatrix}, \quad B_{aug} = \begin{bmatrix} B \\ 0 \end{bmatrix} \quad (18)$$

The controllability matrix of the augmented system presented rank 5, confirming full controllability.

The LQR minimizes the quadratic cost:

$$J = \int_0^\infty (x^T Q x + u^T R u) dt \quad (19)$$

Weighting matrices were selected using Bryson's rule, setting each diagonal entry as the inverse square of the maximum allowable deviation for the corresponding state:

$$Q = \text{diag}\left(\frac{1}{p_{\max}^2}, \frac{1}{\theta_{\max}^2}, \frac{1}{\dot{p}_{\max}^2}, \frac{1}{\dot{\theta}_{\max}^2}, \frac{1}{\xi_{\max}^2}\right) \quad (20)$$

$$R = \frac{1}{u_{\max}^2} \quad (21)$$

The maximum allowable values used were:

TABLE III BRYSON'S RULE DESIGN PARAMETERS			
State	Symbol	Max. value	Q_{ii}
Position	p	0.112 m	80.0
Angle	θ	0.209 rad	23.0
Linear vel.	$\dot{p}$	1.0 m/s	1.0
Angular vel.	$\dot{\theta}$	0.354 rad/s	8.0
Integral error	ξ	0.158 m · s	40.0
Control input	u	1.0 V	$R = 1.0$

The resulting augmented gain vector was:

$$\hat{K}_{LQR} = [80.0354 \ -23.1906 \ 7.1329 \ -7.0107 \ -40.0000] \quad (22)$$

Thus, the state-feedback and integral gains were:

$$K = [80.0354 \ -23.1906 \ 7.1329 \ -7.0107] \quad (23)$$

$$K_I = -40.0 \quad (24)$$

The obtained closed-loop poles were:

$$\lambda_{cl} = \{-10.4662, -1.4987 \pm 6.1233j, -0.7928 \pm 0.4876j\} \quad (25)$$

IV. RESULTS AND ANALYSIS

A. Pole Placement Results

B. LQR Results

The LQR controller was evaluated using a 0.5 m step reference under both simulation and experimental conditions.

1) *Simulation Results:* The following performance metrics were obtained in simulation:

- Settling time: $t_s = 4.701$ s
- Overshoot: $M_p = 0.601\%$
- Zero steady-state error

The simulated cart position response is shown in Fig. 1, and the corresponding pendulum angle response in Fig. 2.

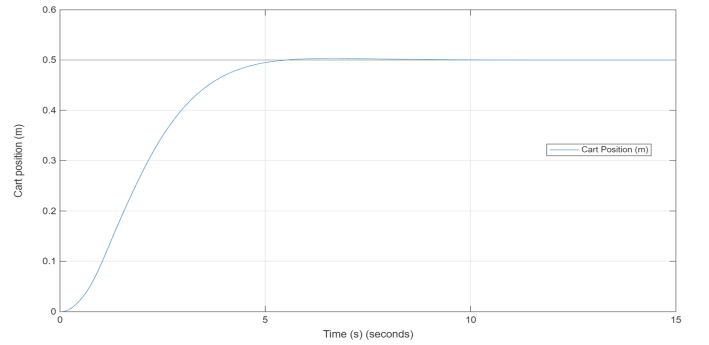


Fig. 1. Simulated cart position $p(t)$ using the LQR controller for a 0.5 m step reference.

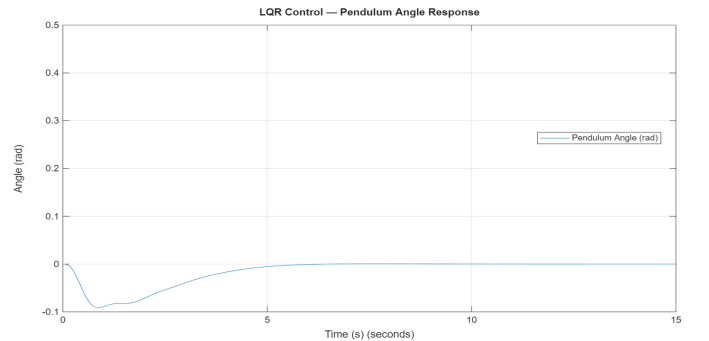


Fig. 2. Simulated pendulum angle $\theta(t)$ using the LQR controller for a 0.5 m step reference.

2) *Experimental Results:* The controller was subsequently implemented on the physical plant. The measured performance metrics were:

- Settling time: $t_s = 5.25$ s
- Overshoot: $M_p = 0\%$
- Zero steady-state error

The experimental cart position response is shown in Fig. 3, and the pendulum angle response in Fig. 4.

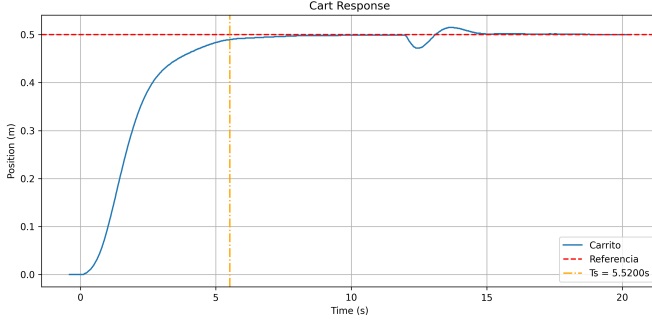


Fig. 3. Experimental cart position $p(t)$ using the LQR controller for a 0.5 m step reference.

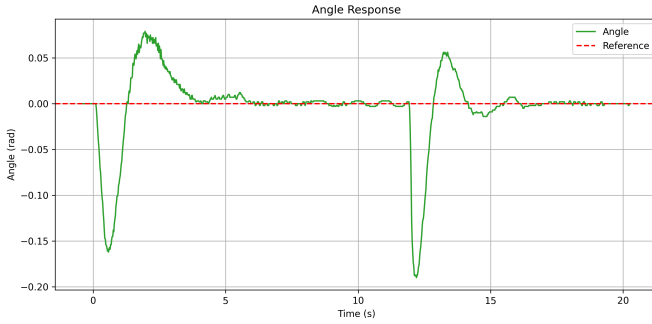


Fig. 4. Experimental pendulum angle $\theta(t)$ using the LQR controller for a 0.5 m step reference.

The experimental results are consistent with the simulated response, with a moderate increase in settling time attributable to nonlinear effects not captured by the identified linear model, such as friction, motor dead-zone, and actuator saturation. The absence of overshoot in the experimental response suggests that the physical damping exceeds that of the linearized model.

C. Controller Comparison

Table III summarizes the performance metrics of both controllers.

TABLE IV
CONTROLLER PERFORMANCE COMPARISON

Controller	Condition	t_s (s)	OS (%)	e_{ss}
Pole Placement	Simulation	—	—	—
	Experimental	—	—	—
LQR	Simulation	4.701	0.601	0
	Experimental	5.25	0	0

V. DISCUSSION

VI. CONCLUSIONS

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